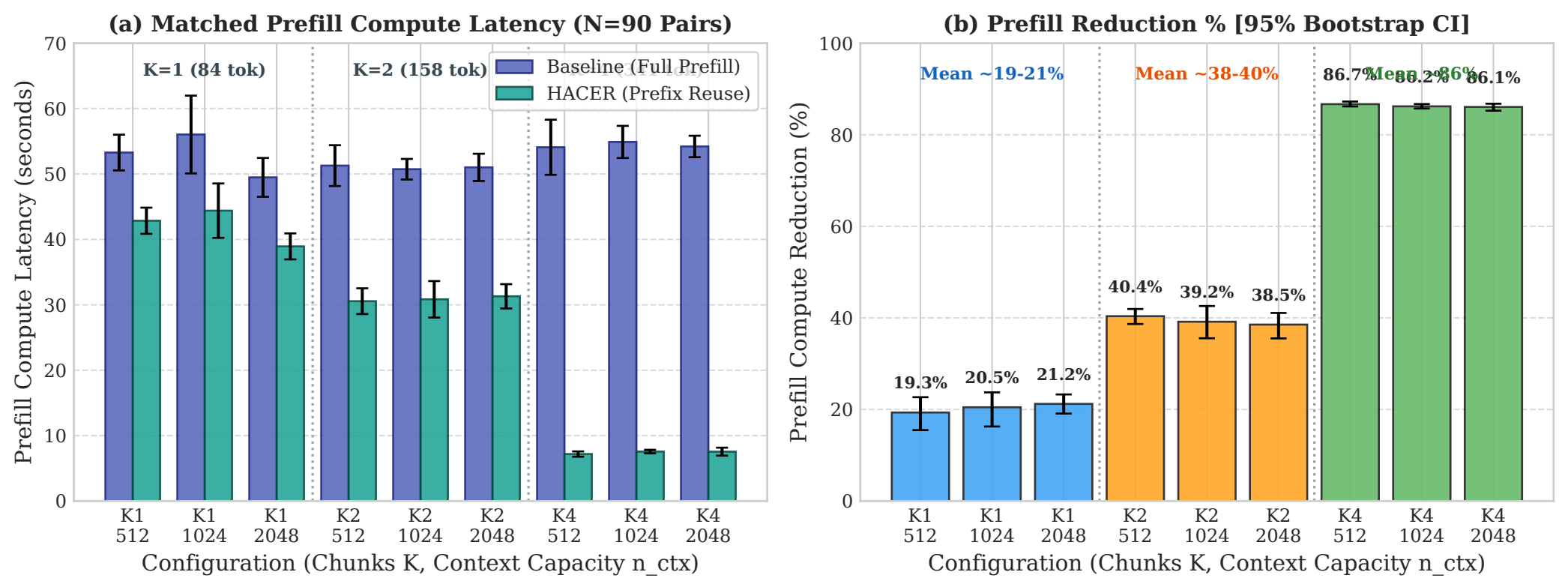
(a) Matched Prefill Compute Latency (N=90 Pairs)
K=1 (84 tok)
K=2 (158 tok)
Baseline (Full Prefill)
HACER (Prefix Reuse)
Prefill Compute Latency (seconds)
0
10
20
30
40
50
60
70
K1 512
K1 1024
K1 2048
K2 512
K2 1024
K2 2048
K4 512
K4 1024
K4 2048
Configuration (Chunks K, Context Capacity n_ctx)
(b) Prefill Reduction % [95% Bootstrap CI]
Mean ~19-21%
Mean ~38-40%
Mean ~86%
19.3%
20.5%
21.2%
40.4%
39.2%
38.5%
86.7%
86.2%
86.1%
Prefill Compute Reduction (%)
0
20
40
60
80
100
K1 512
K1 1024
K1 2048
K2 512
K2 1024
K2 2048
K4 512
K4 1024
K4 2048
Configuration (Chunks K, Context Capacity n_ctx)